

to reduce friction. They applied ultrasonic vibrations to the plate—known to reduce friction between the fingertips and plate. The finger motion during the vibration was measured and contact between the fingertips and the plate imaged. In order to avoid the moisture problem with human fingertips, the researchers also built artificial fingertips constructed of aluminium cylinder representing the bone and a sponge-like material to represent the soft tissue and repeated the experiments.

For ultrasonic vibration that moves the glass surface as little as about 3 micrometres—nearly 40 times less than the width of a human hair—the scientists measured nearly 95 per cent reduction in friction force when a fingertip swiped the glass plate. And the reason for the reduction was a cushion of air trapped between the finger and the glass surface that bolstered these bounces preventing them from going out of control.

Together these results show the need for a bouncing skin motion and compressed air between the fingertips and the surface that will be successful in reducing friction. This friction reduction would play a key role in designing smart devices that provide touch feedback.

Modified, 3D-printable alloy shows promise for flexible electronics

Oregon State University (OSU) researchers believe that their work in 3D printing tall (up to 10mm high and 20mm wide), complex structures with a highly conductive gallium alloy may be the key to the rapid manufacture of flexible computer screens and other stretchable electronic devices.

The team put nickel nanoparticles into the liquid metal, galinstan, to thicken it into a paste with a consistency suitable for additive manufacturing.

“The runny alloy was impossible to layer into tall structures,” explains Yiit Mengüç, assistant professor of mechanical engineering at OSU. “With the paste-like texture, it can be layered while maintaining its capacity to flow, and to stretch inside of rubber tubes. We demonstrated the potential of our discovery by 3D printing a very stretchy two-layered circuit, where the layers weaved in and out of each other without touching.”

Additive manufacturing enables fast fabrication of intricate designs and circuitry. Researchers believe this system could show promise for electrically conductive textiles, bendable displays, wearable sensor suits and biomedical sensors.

Smart kitchen hub to keep an eye on things

GE Appliances is launching an over-the-range smart kitchen hub. The 68cm (27-inch) connected display allows users to access calendars and recipes, and has two cameras: one for live video chat and the other facing the cooktop to monitor food and share images on social media.

The unit, with its stainless steel or black slate frame, resembles the face of a microwave oven. But taking a cue from Samsung's Family Hub fridge, the connected display allows



If the kitchen is the heart of the home, the range is the heart of the kitchen, says GE Appliances, making it the perfect spot for the company's new OTR smart hub

users to access recipes and appointment calendars, and receive over-the-air software updates. Unlike the refrigerator though, whose cameras are aimed inside, the GE hub's forward-facing lens provides multiple angles for live video chat, while a cooktop-facing shooter and onboard lighting lets home chefs share their culinary accomplishments on social media.

The device, enabled by parent company Haier Group's U+ Smart Life platform, also allows the kitchen-bound to see who's ringing the smart doorbell without leaving the room, and to connect and communicate with other smart GE appliances, including washers, dryers, dishwashers, refrigerators, ranges and wall ovens.

High-tech, yet comfortable clothing

Uncomfortable, rigid, with low air permeability—textile materials capable of conducting electricity can be awkward for day-to-day use. However, researchers at the University of Bayreuth, Donghua University in Shanghai, and Nanjing Forestry University have now developed new nonwoven materials that are electrically conductive as well as flexible and breathable. This paves the way for comfortable high-tech clothes that, for example, convert sunlight into warmth, supply wearable electronic devices with electricity, or contain sensors for fitness training.